A Coin Game

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1 Proposed Problem

Problem. Proposed by Dave Kane (Student).

Two players each flip their own coin F times. Player 1 got heads on each flip with success probability p_1, and Player 2 with success probability p_2. What is the probability $P_{\text{tie}}(F)$ that both players end up with the exact same number of heads? Completely characterize both the exact closed-form representations and the asymptotic decay behavior of $P_{\text{tie}}(F)$ as $F \to \infty$ for all configurations of p_1 and p_2.

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2 Solution

Since the coins are flipped independently, the probability of a tie is the sum of the joint probabilities of obtaining exactly k heads:

$$P_{\text{tie}}(F) = \sum_{k=0}^{F} \binom{F}{k}^2 (p_1 p_2)^k [(1-p_1)(1-p_2)]^{F-k}.$$

We analyze this tie probability by partitioning the parameter space into three progressive regimes.

2.1 Case 1: The Fair Coin ($p_1 = p_2 = 1/2$)

For fair coins, the success and failure probabilities are identical:

$$(p_1 p_2)^k[(1-p_1)(1-p_2)]^{F-k} = \left(\frac{1}{4}\right)^k \left(\frac{1}{4}\right)^{F-k} = 4^{-F}.$$

Since this term is independent of k, we factor it out of the sum:

$$P_{\text{tie}}(F) = 4^{-F} \sum_{k=0}^{F} \binom{F}{k}^2.$$

To simplify the sum of squared binomial coefficients, we apply **Vandermonde's Identity**:

$$\sum_{k=0}^{r} \binom{m}{k}\binom{n}{r-k} = \binom{m+n}{r}.$$

Setting $m = n = r = F$ and using the symmetry property $\binom{F}{k} = \binom{F}{F-k}$:

$$\sum_{k=0}^{F} \binom{F}{k}^2 = \sum_{k=0}^{F} \binom{F}{k}\binom{F}{F-k} = \binom{2F}{F}.$$

Substituting this back yields the exact, closed-form baseline:

$$P_{\text{tie}}(F) = 4^{-F}\binom{2F}{F}.$$

2.1.a Large-F Asymptotic Approximation

To find the asymptotic behavior, we apply **Stirling's approximation**, $n! \approx \sqrt{2\pi n}\,(n/e)^n$, to both factorials in the binomial coefficient $\binom{2F}{F} = \frac{(2F)!}{(F!)^2}$:

$$(2F)! \approx \sqrt{4\pi F}\,\frac{(2F)^{2F}}{e^{2F}}, \qquad (F!)^2 \approx 2\pi F\,\frac{F^{2F}}{e^{2F}}.$$

Dividing these expressions:

$$\binom{2F}{F} \approx \frac{\sqrt{4\pi F}\cdot (2F)^{2F}/e^{2F}}{2\pi F \cdot F^{2F}/e^{2F}} = \frac{2\sqrt{\pi F}\cdot 4^F F^{2F}}{2\pi F\cdot F^{2F}} = \frac{4^F}{\sqrt{\pi F}}.$$

Substituting this back cancels the 4^{-F} term, revealing a polynomial decay of order $O(F^{-1/2})$:

$$P_{\text{tie}}(F) \sim \frac{1}{\sqrt{\pi F}}.$$

2.1.b Local Limit Theorem Justification of the $\sqrt{\pi}$ Prefactor

To understand the origin of the $\sqrt{\pi}$ term rigorously, we apply the **de Moivre-Laplace Local Limit Theorem (LLT)**. Let S_F be the number of heads, which has mean $\mu_F = F/2$ and variance $F\sigma^2 = F/4$. The LLT states that:

$$\Pr(S_F = k) = \frac{1}{\sqrt{2\pi F\sigma^2}} e^{-\frac{(k-F/2)^2}{2F\sigma^2}} + o\left(\frac{1}{\sqrt{F}}\right).$$

Since the support of S_F consists of integers with lattice spacing $d = 1$, we can approximate the sum of squared probabilities as a Riemann sum. Let $\Delta x = \frac{1}{\sqrt{F\sigma^2}}$ and $x_k = \frac{k-F/2}{\sqrt{F\sigma^2}}$. Then:

$$P_{\text{tie}}(F) = \sum_{k=0}^{F} \Pr(S_F = k)^2 \sim \sum_{k=0}^{F} \left(\frac{1}{\sqrt{2\pi F\sigma^2}} e^{-\frac{x_k^2}{2}} \right)^2 = \frac{1}{2\pi F\sigma^2} \sum_{k=0}^{F} e^{-x_k^2}.$$

We factor out one copy of the lattice step size $\Delta x = \frac{1}{\sqrt{F\sigma^2}}$:

$$P_{\text{tie}}(F) \sim \frac{1}{2\pi\sqrt{F\sigma^2}} \sum_{k=0}^{F} e^{-x_k^2} \Delta x.$$

In the limit $F \to \infty$, the Riemann sum converges to the integral of a squared standard normal density:

$$\lim_{F\to\infty} \sum_{k=0}^{F} e^{-x_k^2} \Delta x = \int_{-\infty}^{\infty} e^{-x^2} dx = \sqrt{\pi}.$$

Substituting $\sigma = 1/2$:

$$P_{\text{tie}}(F) \sim \frac{\sqrt{\pi}}{2\pi\sqrt{F\sigma^2}} = \frac{1}{2\sigma\sqrt{\pi F}} = \frac{1}{\sqrt{\pi F}}.$$

This formalizes the "collision probability" heuristic: the $\sqrt{\pi}$ prefactor is the normalization constant of the Gaussian distribution, arising directly from integrating the squared density over a 1D lattice.

2.2 Case 2: Matched but Unfair Coins ($p_1 = p_2 = p \neq 1/2$)

When the players use matching biased coins ($p \neq 1/2$), the term $p^2 \neq (1-p)^2$, which prevents factoring out the probability term from the sum and breaks the simple Vandermonde collapse.

However, because the two probability distributions are centered at the identical mean Fp, their overlap remains maximized. The exact tie probability is represented by the direct sum:

$$P_{\text{tie}}(F) = \sum_{k=0}^{F} \binom{F}{k}^2 p^{2k}(1-p)^{2(F-k)}.$$

Since this sum does not admit a simpler elementary algebraic closed-form representation, we analyze its asymptotic behavior as $F \to \infty$. By applying the de Moivre-Laplace Local Limit Theorem with variance $\sigma^2 = p(1-p)$, the tie probability still decays polynomially as $\Theta(F^{-1/2})$, scaled by the variance of the biased coin:

$$P_{\text{tie}}(F) \sim \frac{1}{2\sqrt{\pi F p(1-p)}}.$$

2.3 Case 3: General Asymmetric Coins ($p_1 \neq p_2$)

When $p_1 \neq p_2$, the two binomial distributions are centered at different expectations Fp_1 and Fp_2. This shifts the overlap of the distributions into their tail regions, driving the tie probability to decay **exponentially**. The exact tie probability is given by the general sum:

$$P_{\text{tie}}(F) = \sum_{k=0}^{F} \binom{F}{k}^2 (p_1 p_2)^k [(1-p_1)(1-p_2)]^{F-k}.$$

2.3.a Exact Analytical Asymptotic Formula via Laplace's Method

We can derive the exact analytical asymptotic formula, including both the exponential rate and the polynomial prefactor, using Laplace's method for sums.

Theorem 2. *Let $p_1 \neq p_2$. Define the positive constants:*

$$A = \sqrt{p_1 p_2}, \qquad B = \sqrt{(1-p_1)(1-p_2)}.$$

The asymptotic tie probability as $F \to \infty$ is:

$$P_{\text{tie}}(F) \sim \frac{A+B}{2\sqrt{\pi F A B}} (A+B)^{2F}.$$

Proof. We rewrite the exact sum by factoring:

$$P_{\text{tie}}(F) = \sum_{k=0}^{F} \left[\binom{F}{k} A^k B^{F-k} \right]^2.$$

Let $f(k) = \binom{F}{k} A^k B^{F-k}$. We wish to find the asymptotic sum $\sum f(k)^2$. Since $A + B = \sqrt{p_1 p_2} + \sqrt{(1-p_1)(1-p_2)} < 1$ for all $p_1 \neq p_2$, the sequence $f(k)$ is not a probability distribution. However, we can express it as a scaled version of a true binomial probability distribution. Specifically, we can write $f(k) = (A+B)^F g(k)$, where $g(k) = \binom{F}{k}(p^*)^k (q^*)^{F-k}$ is a probability distribution with success parameter $p^* = A/(A+B)$ and failure parameter $q^* = B/(A+B)$. This allows us to rigorously apply standard asymptotic results and Stirling's approximation to the peak term of this binomial distribution.

The peak of $f(k)$ occurs at its expectation:

$$k^* = F \cdot p^*, \qquad \text{where} \;\; p^* = \frac{A}{A+B}, \; q^* = 1 - p^* = \frac{B}{A+B}.$$

Applying Stirling's approximation to the peak term $f(k^*)$:

$$\binom{F}{Fp^*} A^{Fp^*} B^{Fq^*} \sim \frac{1}{\sqrt{2\pi F p^* q^*}} (A+B)^F.$$

Near the peak, we write $k = Fp^* + y\sqrt{F}$. The Gaussian approximation of the binomial term yields:

$$\binom{F}{k} A^k B^{F-k} \sim \frac{(A+B)^F}{\sqrt{2\pi F p^* q^*}} \exp\left(-\frac{(k-Fp^*)^2}{2Fp^*q^*}\right).$$

Squaring this term:

$$f(k)^2 \sim \frac{(A+B)^{2F}}{2\pi F p^* q^*} \exp\left(-\frac{(k-Fp^*)^2}{Fp^*q^*}\right).$$

Summing $f(k)^2$ over all k is asymptotically equivalent to integrating this Gaussian profile:

$$\sum_{k=0}^{F} f(k)^2 \sim \int_{-\infty}^{\infty} \frac{(A+B)^{2F}}{2\pi F p^* q^*} \exp\left(-\frac{y^2}{Fp^*q^*}\right) dy.$$

Evaluating this standard Gaussian integral:

$$\int_{-\infty}^{\infty} \exp\left(-\frac{y^2}{Fp^*q^*}\right) dy = \sqrt{\pi F p^* q^*}.$$

Multiplying the prefactors:

$$P_{\text{tie}}(F) \sim \frac{(A+B)^{2F}}{2\pi F p^* q^*} \cdot \sqrt{\pi F p^* q^*} = \frac{(A+B)^{2F}}{2\sqrt{\pi F p^* q^*}}.$$

To simplify, we express p^*q^* in terms of A and B:

$$p^*q^* = \left(\frac{A}{A+B}\right)\left(\frac{B}{A+B}\right) = \frac{AB}{(A+B)^2} \implies \sqrt{p^*q^*} = \frac{\sqrt{AB}}{A+B}.$$

Substituting this into the denominator yields the stated formula:

$$P_{\text{tie}}(F) \sim \frac{(A+B)^{2F}}{2\sqrt{\pi F}\frac{\sqrt{AB}}{A+B}} = \frac{A+B}{2\sqrt{\pi F AB}}(A+B)^{2F}. \qquad \blacksquare$$

Since $A+B = \sqrt{p_1p_2} + \sqrt{(1-p_1)(1-p_2)} < 1$ for all $p_1 \neq p_2$, this proves the exponential decay rate rigorously, while providing the exact polynomial prefactor $\Theta(F^{-1/2})$.

2.3.b Unifying Check on All Three Cases

A beautiful feature of the general asymptotic formula is that it seamlessly unifies all three cases when the parameters are equal ($p_1 = p_2 = p$):

1. **Unfair but Matched Coins ($p_1 = p_2 = p$)**: Setting $p_1 = p_2 = p$ yields $A = p$ and $B = 1 - p$, which gives $A + B = 1$. The exponential decay factor $(A+B)^{2F} = 1^{2F} = 1$ collapses, and the prefactor becomes:

$$P_{\text{tie}}(F) \sim \frac{1}{2\sqrt{\pi F p(1-p)}},$$

recovering the Case 2 polynomial asymptotic decay rate.

2. **Fair Coins ($p = 1/2$)**: Further setting $p = 1/2$ yields $p(1-p) = 1/4 \implies \sqrt{p(1-p)} = 1/2$. The formula reduces to:

$$P_{\text{tie}}(F) \sim \frac{1}{\sqrt{\pi F}},$$

recovering the Case 1 fair-coin baseline.

This provides a complete, elegant, and unified characterization of the coin game tie probabilities under all configurations.

3 References

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